

Towards a Practical Framework for Cryptographic Protocols in Adversarial Cooperation

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Abstract—This study aims to investigate the role of information theory in designing cryptographic protocols in adversarial settings. By leveraging secure multi-party computations; during the research we will build an *open-source* software library demonstrating the results.

I. INTRODUCTION

To de-escalate conflict, we propose to have adversaries cooperate in a structured setting. Where trust is cryptographically enforced. —Implement information-secure and verifiable encryption protocols, securing game-theoretic coordination, and privacy-preserving shared decision-making.

II. LITERATURE REVIEW

Cryptography extends far beyond Encryption, it serves as the foundation of digital trust. The tools at hand for this study are: *Zero-knowledge proofs*, *Homomorphic Computation*, *Garbled Circuits*, *Oblivious Transfer*, *Blockchains*, *Differential Privacy*, and *Threshold Cryptography*; —Combining these secure units enable broad collaborative problem-solving that are believed to be key in adversarial cooperation.

III. RESEARCH QUESTIONS

- 1) What does it mean to apply cryptography to cooperation?
- 2) How can the proposed cryptographic protocols be improved through interdisciplinary collaboration?

IV. RESEARCH OBJECTIVES

- 1) Demonstrate for classical game-theoretic problems.
- 2) Open-source a software library based on this research.

V. METHODOLOGY

The study will focus on the following structured games:

- 1) *Playing card games without revealing one's cards*: —Demonstrate how to claim victory in a card game without exposing the cards at hand.
- 2) *Playing tic-tac-toe without revealing one's strategy*: —Demonstrate how a player can prove they have a winning strategy in tic-tac-toe without disclosing the actual moves.
- 3) *Classical adversarial stag-hunt*: —Demonstrate a path to equilibrium for this fundamental game of adversarial cooperation.

VI. ACCEPTANCE LETTER REQUEST AND JUSTIFICATION

I kindly, respectfully ask the distinguished faculty members consideration and support for my application to *Beihang University* Master's in Cyberspace Security.

Universities in Colombia currently lack research programs in Applied Cryptography. I have been diligently studying the Chinese language for several years, recognizing the importance of seamless communication and collaboration in a Chinese academic setting. I am deeply enthusiastic about the prospect of learning from and contributing to the vibrant research community in China. I believe this scholarship would not only enable me to pursue my professional aspirations but also foster valuable academic exchange between our nations.

VII. EXPECTED OUTCOMES

- 1) Demonstration adversarial cooperation in game-theory.
- 2) Development of an open-source cryptographic protocol suite enabling cooperation in adversarial settings.
- 3) Advocate on how cryptography facilitate trust in conflict.

VIII. CONCLUSION

This research seeks to show how cryptographic techniques can be used for peace-building and trust establishment.

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